
Predicting the win probability using logistic regression for top four English Premier League teams

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Abstract:

Predicting the outcomes of football competitions has long piqued the attention of the general public as well as bookmakers. A combination of algorithm, ranking criteria and scoring scales can be used to decide [the outcome of a game](#). The English Premier League (EPL) is the most famous and watched football league in the world. This research seeks to examine the factors that can affect the winning-probability of top four teams in the running season, 2020-2021. They are Manchester city, Manchester united, Leicester city and Chelsea. The match results data have been collected from these four teams for 105 matches in EPL. This includes the 30 matches of current season, 38 matches of season 2019-2020, and 38 matches of the season 2018-2019. The data are analyzed using the logistic regression implementing the `glm()` function of ISLR package in R. The main findings of this study suggest that the probability of winning a match by Manchester City increases if there are direct free kicks, defending set pieces, creating chances, and attacking down the wing increases. In case of Manchester united, [the win-probability](#) increases if long-shot, counter attacks increase. The winning probability of Manchester united decreases if its offside increases. In case of Leicester city, the winning probability decreases if the opposite team make counter attacks. For the team Chelsea, the free kick shooting increases the winning probability. The Chelsea, however, can make the winning probability decreases by making individual errors.

Keywords: English Premier League, [Football prediction](#), Probability, Betting, [MightyTips](#), Man City, Man United, Leicester City, Chelsea, `glm()`, R.

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Introduction

The English Premier League is the most-watched league on the planet, with one billion homes watching the action in 188 countries. It is home to some of the most popular clubs, teams, coaches, and stadiums in world football. The league lasts from August to May, with teams facing each other at home and away over the course of the season, for a total of 380 games [1]. A victory is worth three points, a draw is worth one point, and a loss is worth zero, with the team with the most points at the end of the season securing the Premier League trophy [2].

The Premier League is the highest tier of England's football pyramid, with 20 clubs vying for the status of English champions. Teams that place first, second, third, and fourth. The lowest three teams in table in the league table at the end of the season are relegated to the England's second tier of football. Such clubs are replaced by three clubs promoted from the Championship: the first and second -placed teams, as well as the third-placed team from the end-of-season playoffs. The Premier League's first season was in 1992/93. Participate in the 22-team league. Brian Dean, a Sheffield United player, scored the first goal in his team's Premiership win, a 2:1 victory over Manchester United. The number of times a Premiership was limited to 20 in 1995 [3]. Despite FIFA's attempts to minimize them, they remain so today, with 18 made. Several attempts have been made to forecast football games using time series evidence, but humans remain superior at predicting sport outcomes. There are a number of commercial services that specialize

in sports research and forecast. They use “advanced tools and mathematical algorithms” to help them monitor data, but they also have experts personally reviewing the games [4].

Human and algorithm forecasts both face difficulties. Humans are human beings with emotions, and their emotions will influence team and, as a result, prediction. A computer may not provide access to information about the team's current mental wellbeing. The result may be affected by rifts between players and coaches. The problem of determining which attributes are important is a problem that both humans and computers face.

Football, like all other disciplines, is somewhat unpredictable. From hundreds of throws, shoots, and dribbles, one lucky strike will ultimately change the game's outcome. This makes it more difficult for humans and computers to forecast the results of football matches [5].

Many people thought that predicting the result of football matches was futile before the 1990s because it was based on chance. Stuart Coles and Mark Dixon's study, on the other hand, changed everything. The team used the 'Poisson Method,' which was named after Simeon Poisson, a physicist.

When you think anything fits this method, you're assuming that things happen at a certain time. After all, what happens in the past doesn't actually determine what will happen in the future. A game with no goals in the first half is no more likely than one with at least one score to have goals in the second half. As a result, the original Dixon and Coles model assumed that goals were scored at a constant and continuous pace during a game. They also believed that the overall number of goals differed based on the teams involved. They wanted to figure out how many goals each team should hope to score [6]. In the end, Dixon and Coles split the team into two groups: attack and defending. The home team's predicted goal total was determined by:

*Their offensive potential * The away team's defensive vulnerability * Home advantage*

Away teams' expected goals were determined by:

*Their offensive potential * The home team's defensive vulnerability*

The research by [7] had data for more home losses (36) than home victories (36) in the last nine rounds without an audience (27). As a result, the Covid-19 lock-down placed the team at a disadvantage at home. One explanation for this unexpected finding may be that the home side is lacking a key familiarity factor while playing in front of an empty stadium with little social support. In addition, since all sides are aware of the HA, the away team might be more inspired in this rare scenario.

In general, home teams win more games than away teams. The support of local fans is often quoted as a reason. The disparity in the percentage of home wins before and after the COVID-19 pandemic in 63 leagues around the world is discussed in issue 304 of the CIES Football Observatory Weekly Post. It shows that the home advantage continued in the absence of fans, although to a lesser extent.

Using the outcomes of these closed matches, the key goal of this analysis by [8] is to do a comparative assessment of 'crowd influence' on home advantage. To mitigate the consequences of the unbalanced timetable, the proposed study employs the pairwise comparison approach. The statistical hypothesis experiments performed in this study led to the following conclusions: In four major European leagues, the home advantage is lower in closed matches than in open matches, i.e. where there are no fans. The rates of reduction varied between leagues. For example, during the closed-match era in Germany, the home advantage was negative. In England, on the other hand, statistically relevant. In England, however, statistically important variations in home advantage between closed matches and usual conditions were not found.

Literature review

Over the last few years, various ranking methods and their adaptations to various sports have been well known. The general population, as well as bookmakers, have long been interested in predicting the results of sporting events. The result of a game can be determined using a variety of ranking criteria and rating systems.

A Brownian motion model can be used to examine how a team's chances of winning change as the game progresses. This model considers the distance by which a home team leads or trails, as well as the remaining time in the match. This model was tested on 493 professional basketball games, with the scores at the conclusion of each game taken into account [9].

A Brownian motion model can be used to examine how a team's chances of winning change as the game progresses. This model considers the distance by which a home team leads or trails, as well as the remaining time in the match. This model was extended to 493 professional basketball games, and it was concluded that the Brownian motion model offered a decent fit to the results by taking into account the scores at the end of each quarter.

When evaluating the rankings of a sport, logistic regression models are often used. The formula was introduced to college football teams by Lebovic and Sigelman [10] to assess the amount of points a team goes up or down in the rankings.

The formula was introduced to college football teams by Lebovic and Sigelman in order to assess the amount of points a team goes up or down in the rankings from week to week [10]. The findings of this model found that a team is more likely to go up in the rankings if they beat a higher-ranked opponent, and a team is more likely to slip in the rankings if they lose to a lower-ranked opponent. Statistical forecasts also struggle to make better than fair estimates than experts of a particular sport. The game predictions of 496 NFL matches were matched by Boulier et al. [11]. Both mathematical forecasts and professional football analysts made these forecasts.

These forecasts were compared to one another as well as to the betting line's predictions. The histories of the players, points scored, yards gained, home field advantage, and other variables were used as input to the mathematical models. Although neither was able to beat the betting line's forecasts, the analysts' predictions were superior to the mathematical models' predictions. A research performed by Boulier and

Stekler in regards to forecasting the outcomes of National Football League matches [11] yielded another finding in which the betting market appeared to be the best predictor of match performance. They made use of the power scores that were produced.

They developed probit regression models using the power scores produced by the New York Times. The predictions made by these models were compared to predictions made by models focused on the betting market and sports editors' opinions. The models focused on the betting market were found to be the best at forecasting the results of National Football League matches, whilst the probit regression models did marginally better than the forecasts of sports editors.

Boulier et al. use Cohen's kappa coefficient to determine the degree of consensus between two factors, in this case, football experts and statistical systems, to test the forecasting ability of National Football League matches [12]. Using Cohen's kappa coefficient, it was determined that mathematical systems had a higher degree of consensus than football experts. The literature contains a wealth of essential principles and techniques. Both logistic regression and ordered probit models may use Elo ratings as data. These versions can be used in a wide range of sports, including soccer.

It's also crucial to consider how a team's probability of winning changes as the game continues. This definition can be examined using a variety of game statistics, and this theory will be examined and applied in this research.

Methodology

In this class of models, the dependent variable, can take on only two values. Y might be a dummy variable reflecting the occurrence of a case, or an option between two alternatives. For example, the results of each match of league sample may be interesting in modeling (whether won or not). The teams vary in

many measurable features, which we call x . The objective is to quantify the connection between team features and the likelihood of winning the game.

Binary variable dependency, y , which takes zero and one values. It is not sufficient to simply linearly regress y on x , as the implied model of the conditional average puts inadequate limitations on residuals of the model. In addition, the value of y from a simple linear regression is not limited to zero [13].

Instead, we follow a specification to deal with the basic needs of binary dependent variables. Suppose we model the likelihood that one is observed as:

$$\Pr(y_i = 1 | x_i, \beta) = 1 - F(-x_i' \beta),$$

Where, F is a continuous function that is purely increasing and takes a value that is true and returns a value of zero to one. What type of binary model would be selected is determined by the choice of function F . It follows that:

$$\Pr(y_i = 0 | x_i, \beta) = F(-x_i' \beta).$$

We may use the maximum likelihood approach to approximate the parameters of this model given such a specification [14]. The likelihoods function is shown as:

$$l(\beta) = \sum_{i=0}^n y_i \log(1 - F(-x_i' \beta)) + (1 - y_i) \log(F(-x_i' \beta)).$$

Since the first order conditions for this probability are nonlinear, an iterative solution is needed to obtain parameter estimates.

This specification has two different views that are worth considering. First, the binary model is often used to specify latent variables. Assume that there is a latent variable y^* that is linearly related to x but not observed.

$$y_i^* = x_i' \beta + u_i$$

Where, u denotes a random fluctuations y^* then determines if the observed dependent variable reaches a threshold value:

$$y_i = \begin{cases} 1 & \text{if } y_i^* > 0 \\ 0 & \text{if } y_i^* \leq 0. \end{cases}$$

The threshold is set to zero in this situation, but the amount of the threshold is meaningless as long as x contains a constant term. Then:

$$\Pr(y_i = 1 | x_i, \beta) = \Pr(y_i^* > 0) = \Pr(x_i' \beta + u_i > 0) = 1 - F_u(-x_i' \beta)$$

Where F_u is the cumulative distribution function of u .

The constraint of coding y as 0 and 1 has some benefits. For one thing, this coding means that the predicted value of y is actually the probability that $y=1$:

$$\begin{aligned} E(y_i | x_i, \beta) &= 1 \cdot \Pr(y_i = 1 | x_i, \beta) + 0 \cdot \Pr(y_i = 0 | x_i, \beta) \\ &= \Pr(y_i = 1 | x_i, \beta). \end{aligned}$$

The convention gives one a second understanding of the binary specification: as a conditional mean.

The assumption that estimated coefficients from a binary model cannot be viewed as the residual effect on the dependent variable complicates interpretation of the coefficient values.

The marginal effect of an independent x variable on one conditional probability of the dependent variable y is calculated as: [15]

$$\frac{\partial E(y_i | x_i, \beta)}{\partial x_{ij}} = f(-x_i' \beta) \beta_j.$$

The list of independent variables:

Shooting from direct free kick, Finishing scoring chances, Creating chances through individual skill, Defending set pieces, creating scoring chances, attacking down the wings, Creating long shot opportunities, Coming back from losing positions, counter attacks, Getting back the ball from the opposition.

Table 1 displays the Summary statistics of the dependent variable for each team (For last 105 matches) for seasons, 2020-2021 (Upto 6th April, 2021), 2019-2020, and 2018-2019

Table 1:

Team	win	loss	draw
Manchester City	23+26+32 =81	3+9+4 =16	5+3+2 =10
Manchester United	19+18+17 =54	10+8+4 =22	9+12+9 30
Leicester City	17+18+15 =50	8+12+16 =36	5+8+7 =20

Chelsea	14+20+21 =55	7+12+8 =27	9+6+9 24
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Results:

The table 2, table 3, table 4, and table 5 displays the results of logistic regression for each team. The summary of the results are reported in tables (6-9) with extra control variables.

Table 2. Manchester City

Dependent Variable: Match results

Method: ML - Binary Logit (BFGS / Marquardt steps)

Sample: 1 105

Included observations: 105

Convergence achieved after 24 iterations

Coefficient covariance computed using observed Hessian

Variable	Coefficient	Std. Error	z-Statistic	Prob.
Direct Free Kick	-13.02135	4.9311054	-2.640538	0.0083
Creating Chances (Individual)	2.826113	1.262941	2.237723	0.0252
Defending Set Pieces	0.095158	0.141554	0.672235	0.5014
Creating chances	0.378688	1.064564	2.234424	0.0755
Attacking down the wing	0.254657	0.436474	0.734646	0.0964
McFadden R-squared	0.374038	Mean dependent var		0.343750
S.D. dependent var	0.482559	S.E. of regression		0.384716
Akaike info criterion	1.055602	Sum squared resid		4.144171
Schwarz criterion	1.238819	Log likelihood		-12.88963
Hannan-Quinn criter.	1.116333	Deviance		25.77927
Restr. deviance	41.18346	Restr. log likelihood		-20.59173
LR statistic	15.40419	Avg. log likelihood		-0.402801
Prob(LR statistic)	0.001502			

Table 3: Manchester United

Dependent Variable: MATCH RESULTS

Method: ML - Binary Logit (BFGS / Marquardt steps)

Sample: 1 105

Included observations: 105
 Convergence achieved after 28 iterations
 Coefficient covariance computed using observed Hessian

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	-12.49554	5.024561	-2.486891	0.0129
Attacking down the wing	3.245413	1.317937	2.462495	0.0138
Long shot opportunities	0.051144	0.155100	0.1059749	0.7416
Chances through balls	2.563992	1.151528	2.226600	0.0260
Counter attcaks	-2.177585	1.840997	-1.182829	0.2369
offside	-5.343484	4.685747	-3.574746	0.0736
Defending chances by Opponents	0.647766	3.644786	2.546644	0.2345
McFadden R-squared	0.411722	Mean dependent var		0.343750
S.D. dependent var	0.482559	S.E. of regression		0.381145
Akaike info criterion	1.069604	Sum squared resid		3.922338
Schwarz criterion	1.298626	Log likelihood		-12.11367
Hannan-Quinn criter.	1.145519	Deviance		24.22734
Restr. deviance	41.18346	Restr. log likelihood		-20.59173
LR statistic	16.95612	Avg. log likelihood		-0.378552
Prob(LR statistic)	0.001971			

Table 4. Leicester City

Dependent Variable: MATCH RESULTS
 Method: ML - Binary Logit (BFGS / Marquardt steps)

Sample: 1 105
 Included observations: 105
 Convergence achieved after 29 iterations
 Coefficient covariance computed using observed Hessian

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	-17.33966	7.045646	-2.461047	0.0139
Through Ball chances	3.551390	1.621506	2.190181	0.0285
Long shot opportunities	0.115266	0.142913	0.806546	0.4199
Defending set pieces	2.697550	1.148614	2.348525	0.0988
Counter attack of opponents	-2.1058895	2.176724	1.069908	0.0847
Through Ball defense				
McFadden R-squared	0.404426	Mean dependent var		0.343750
S.D. dependent var	0.482559	S.E. of regression		0.374558
Akaike info criterion	1.078994	Sum squared resid		3.787939
Schwarz criterion	1.308015	Log likelihood		-12.26390
Hannan-Quinn criter.	1.154908	Deviance		24.52781
Restr. deviance	41.18346	Restr. log likelihood		-20.59173
LR statistic	16.65565	Avg. log likelihood		-0.3810547
Prob(LR statistic)	0.002255			

Table 5. Chelsea

Dependent Variable: MATCH RESULTS

Method: ML - Binary Logit (BFGS / Marquardt steps)

Sample: 1 105

Included observations: 105

Convergence achieved after 28 iterations

Coefficient covariance computed using observed Hessian

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	-12.74684	5.087821	-2.505364	0.0122
Free kick shooting	2.779547	1.283385	2.165794	0.0303
Attacking set pieces	0.095941	0.140888	0.680975	0.4959
Defending set pieces	2.422588	1.093421	2.215604	0.0267
Individual players errors	-0.371622	1.817542	-0.204464	0.8380
Possession	4.363728	3.474733	6.383727	0.1637
McFadden R-squared	0.375056	Mean dependent var		0.343750
S.D. dependent var	0.482559	S.E. of regression		0.389334
Akaike info criterion	1.116793	Sum squared resid		4.092678
Schwarz criterion	1.345814	Log likelihood		-12.86868
Hannan-Quinn criter.	1.192707	Deviance		25.73736
Restr. deviance	41.18346	Restr. log likelihood		-20.59173
LR statistic	15.44610	Avg. log likelihood		-0.402146
Prob(LR statistic)	0.003860			

Table 6. Results summary for Manchester city

Manchester city

Factors	Level of probability
Shooting from direct free kick	High probability of winning
Finishing scoring chances	High probability of winning
Creating chances through individual skill	High probability of winning
Defending set pieces	High probability of winning
Creating scoring chances	Moderate probability of winning
Attacking down the wings	Moderate probability of winning

Table 7. Results summary for Manchester United

Factors	Level of probability
Finishing scoring chances	High probability of winning
Attacking down the wings	High probability of winning
Creating long shot opportunities	High probability of winning
Creating chances using through balls	High probability of winning
Coming back from losing positions	High probability of winning
Counter attacks	Moderate probability of winning
Creating scoring chances	Moderate probability of winning
Avoiding offside	Weak probability of winning
Stopping opponents from creating chances	Weak probability of winning
Protecting the lead	Weak probability of winning

Table 8. Results summary for Leicester City

Factors	Probability levels
Creating chances using through balls	High probability of winning
Creating long shot opportunities	Moderate probability of winning
Coming back from losing positions	Moderate probability of winning
Shooting from direct free kicks	Moderate probability of winning
Finishing scoring chances	Moderate probability of winning
Protecting the lead	Moderate probability of winning
Getting back the ball from the opposition	Moderate probability of winning
Defending counter attacks	Weak probability of winning
Defending set pieces	Weak probability of winning
Defending against through ball attacks	Weak probability of winning

Table 9. Results summary for Chelsea

Factors	Probability levels
shooting from direct free kicks	High probability of winning
Attacking set pieces	Moderate probability of winning
Coming back from losing positions	Moderate probability of winning
Defending set pieces	Moderate probability of winning
Getting back the ball from the opposition	Moderate probability of winning
individual errors	Weak probability of winning

Conclusion:

The aim of this study is to look into the factors that can influence the top four teams' chances of winning in the current season, 2020-2021 for the English Premier League. Manchester City, Manchester United, Leicester City, and Chelsea are the squads. For 105 EPL matches, data on match results was obtained from these four teams. This includes the current season's 30 matches, the 2019-2020 season's 38 matches, and the 2018-2019 season's 38 matches. The data is analyzed using logistic regression, which is implemented in R using the `glm()` function from the ISLR package. The study's key results indicate that direct free kicks, defending set pieces, and creating chances improve Manchester City's chances of winning a match.

The key results of this study indicate that direct free kicks, defending set pieces, creating opportunities, and attacking down the wing all improve Manchester City's chances of winning a match. If long-shot, counter-attacks increase, Manchester United's chances of winning increase. Manchester United's chances of winning go down as the offside percentage rises. If the opposing team makes counter attacks, Leicester City's chances of winning decrease. Chelsea's chances of winning rise as a result of their free kick shooting. Chelsea, on the other hand, will reduce their chances of winning by making individual errors.

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